

E Taxpay

Multilingual Shop Tax Management System for Uttarakhand Zila Panchayat

*A project Report submitted in partial fulfilment of the
requirements for the award of the diploma of*

Computer Science

(POLYTECHNIC)

Submitted by

Sumit Bhandari (23060120015)

Manish Paliwal(23009050012)

Bhavesh Bisht (23009050004)

Raja Rautela (23009050018)

Lalit Singh Kanyal(23009050034)

Sahil Chand (23009050014)

Deepak Singh Bisht (23050050001)

Gaurav Bisht(23009050020)



Under the supervision of

Mr.Manmohan sir
Lecturer C.S.E
C.S.E Department

Mr.Virendra sir
Lecturer C.S.E
C.S.E Department

**DEPARTMENT OF COMPUTER SCIENCE
GOVT. POLYTECHNIC KASHIPUR
KASHIPUR -244713 (UTTARAKHAND) INDIA**

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Sumit Bhandari
Manish Paliwal
Sahil Chand
Lalit Singh Kanyal
Deepak Bisht
Raja Rautela
Bhavesh Bisht
Gaurav Bisht

Candidate's Declaration

I/We hereby declare that the project titled **“E TaxPay – Multilingual Shop Tax Management System for Uttarakhand Zila Panchayat”** submitted to the Government Polytechnic Kashipur in partial fulfillment of the requirements for the Diploma in Computer Science and Engineering is an original work carried out by us under the guidance of Mr. Manmohan. No part of this report has been submitted previously for any other diploma or degree. All sources of information and external contributions are appropriately cited in the references.

Date:

Place: Kashipur

Sumit Bhandari
Manish Paliwal
Sahil Chand
Lalit Singh Kanyal
Deepak Bisht
Raja Rautela
Bhavesh Bisht
Gaurav Bisht

SUPERVISOR CERTIFICATE

This is to certify that the project report titled “**E TaxPay – Multilingual Shop Tax Management System for Uttarakhand Zila Panchayat**” submitted by the students listed above to the Government Polytechnic Kashipur is a record of original work carried out by them under my supervision. To the best of my knowledge, the work is original, and all sources have been cited.

Mr. Manmohan
Head Of Department CSE

Mr. Virendra Dharamsaktu
Lecturer CSE

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CHAPTER 1: INTRODUCTION

1.1 Need of the Study

In the current administrative setup, local governing bodies such as the Zila Panchayat face significant challenges in managing and collecting taxes from businesses and property owners. Traditional methods predominantly involve manual record-keeping, reliance on physical paperwork, and cash transactions. These antiquated systems are not only prone to human error and inefficiencies but also open up numerous avenues for verification fraud and tax evasion.

Furthermore, taxpayers often experience long waiting times, bureaucratic hurdles, and a lack of transparency regarding how their taxes are calculated and where their payments are recorded. The manual reconciliation of these financial records at the end of the fiscal year is a tedious and labor-intensive process for municipal workers, leading to delayed audits and financial discrepancies.

Therefore, there is a pressing need for a modernized, transparent, and centralized digital platform to streamline tax administration. Implementing an e-governance system like E-TaxPay can ensure data integrity, facilitate real-time tracking of tax dues, and significantly enhance convenience for rural and semi-urban populations by allowing them to clear their dues from the comfort of their homes. This digital transition aligns with the broader goals of establishing a transparent and accountable public administration.

1.2 Applications of the Study

The E-TaxPay platform is designed as a dual-facing system, serving both the administrative officials and the general public, thereby acting as a bridge between the government and its citizens.

1. Zila Panchayat Authorities (Administrators):

Administrators gain access to a centralized, real-time dashboard that provides a comprehensive overview of tax collection metrics. They can monitor incoming payments, verify submitted business credentials, track defaulters, and generate detailed analytical reports. The system allows them to easily cross-reference uploaded documents, such as rental agreements and utility bills, to ensure the authenticity of the taxpayers and their properties.

2. Citizens and Business Owners (Taxpayers):

For taxpayers, the application serves as a self-service portal. Users can register their properties, calculate their exact tax liabilities based on current civic rules, make secure online payments, and download digital payment receipts instantly.

Additionally, recognizing the demographic diversity of the region, the platform incorporates multi-language support (English and Hindi). Alongside the web portal, a dedicated ****Mobile Application**** has been developed for the E-TaxPay platform. This dual-ecosystem (Web and Mobile App) paired with Progressive Web App (PWA) architecture ensures that users with varying levels of technical proficiency, low-end devices, or poor internet connectivity can access the platform seamlessly, making the system highly inclusive.

1.3 Scope of the Study

The scope of the E-TaxPay platform covers the end-to-end digitalization of the tax management lifecycle specifically tailored for the local government level. The study encompasses the design and

deployment of several core modules:

- **Cross-Platform Ecosystem:** Development of both a comprehensive web portal and a dedicated mobile application to maximize public reach and accessibility.
- **User Management & Authentication:** Establishing secure login protocols using modern cryptographic standards to protect citizen data.
- **Tax Assessment Engine:** An automated calculator that dynamically determines tax dues based on property dimensions, business types, and location zones.
- **Fraud Prevention & Verification:** A robust, multi-tier verification module that prevents attribute fraud by mandating and cross-referencing document uploads prior to account approval.
- **Payment & Receipt Generation:** Secure integration for recording financial transactions alongside an automated PDF receipt generator that provides taxpayers with immediate proof of payment.

Furthermore, the study explores the implementation of Role-Based Access Control (RBAC) to securely differentiate the capabilities of regular citizens, data-entry operators, and high-level administrative officials, ensuring that sensitive data is only accessible to authorized personnel.

1.4 Objective of the Study

The overarching aim of this project is to create an efficient, scalable, and user-friendly digital taxation ecosystem. The primary objectives are:

- To design and develop a full-stack, secure web application and a companion **Mobile Application** ("E-TaxPay") that digitizes the entire tax collection process for the Zila Panchayat.
- To implement a reliable document verification system that actively prevents business and property misrepresentation fraud through structured admin audits.
- To provide an inclusive, accessible user experience by integrating bilingual UI capabilities (English/Hindi) across both mobile and web platforms.
- To automate tedious administrative workflows, including record-keeping, dynamic tax assessment, and financial report generation, thereby drastically reducing the operational overhead and manual labor required by municipal authorities.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The rapid growth of digital technologies has significantly transformed the public administration and e-governance sectors. Traditional methods of managing municipal taxes and property registrations are gradually being replaced by web-based and app-based that offer centralized access, automation, and improved user transparency.

This chapter reviews existing systems and technologies related to zila panchyat tax collection, fraud prevention, cloud databases, and accessible mobile platforms, which form the foundation for the development of the E-TaxPay platform.

2.2 Existing Tax Collection Systems

Many local bodies, such as the Zila Panchayat, still rely on traditional or partially digitized systems for managing tax activities.

Features of Existing Systems:

- Manual ledger entries and paper-based record keeping
- Physical document submission for property verification
- In-person cash or check payments at municipal offices
- Manual calculation of tax dues based on static rules

Limitations:

- Highly prone to human error and data loss
- Rampant verification fraud (e.g., misrepresenting property size)
- Lack of transparency for the taxpayers
- Inefficient processing times and long queues at offices

These limitations highlight the pressing need for a more comprehensive, automated system like E-TaxPay.

2.3 E-Governance and Digital Taxation Platforms

Some state governments provide digital taxation portals to help citizens pay their dues online.

Observations:

- Provide online payment gateways
- Basic database integration for property tracking

Limitations:

- Often lack robust verification mechanisms, allowing fake registrations
- Poor user interfaces that are not optimized for mobile devices
- Mostly available only in English, creating language barriers for rural users

E-TaxPay improves upon this by combining rigorous document verification with a highly accessible, multi-lingual interface.

2.4 Document Verification and Fraud Prevention

Modern financial platforms utilize digital verification to combat identity and attribute fraud.

Advantages of Digital Verification:

- Ensures data authenticity using document uploads (e.g., rental agreements)
- Reduces reliance on physical site inspections
- Maintains an immutable audit trail

Limitations in Current Civic Systems:

- Verification is often a one-time process, leading to outdated records
- Lack of dedicated admin dashboards for efficient auditing

E-TaxPay uses a structured upload and admin-approval pipeline, coupled with periodic re-verification protocols, ensuring data remains accurate over time.

2.5 Cloud-Based Backend Systems

Modern web applications use cloud platforms for scalable database management, authentication, and secure storage.

Platforms like Supabase provide:

- Real-time PostgreSQL databases
- Secure authentication systems
- File storage for uploaded documents
- Seamless API integration

****Benefits:****

- Highly scalable and secure
- Reduces backend development and maintenance effort
- Supports rapid modern web application deployment

E-TaxPay leverages Supabase for managing structured taxpayer data and securely storing verification documents like PDFs and images.

2.6 Progressive Web Apps (PWA) and Localization

Accessibility is critical for public administration platforms. PWAs and multi-language support bridge the digital divide.

Applications:

- App-like experience in standard mobile browsers
- Dynamic toggling between regional languages (Hindi/English)

Advantages:

- No need to download heavy apps from app stores
- Works efficiently on low-end devices and poor networks
- Increases adoption rates among rural and semi-urban demographics

E-TaxPay integrates a PWA architecture and `i18next` localization to ensure every citizen can comfortably navigate the tax portal.

2.7 Gap Analysis

From the above study, the following gaps are identified in existing traditional and digital systems:

- No unified platform combining robust fraud prevention with accessible tax payment
- Absence of localized, multi-language support tailored for rural populations

- Lack of mobile-first architecture (PWAs) in standard municipal portals
- Inefficient manual workflows for administrators to audit and verify taxpayers

Table 2.1: Comparison of Traditional vs. Digital Tax Collection Systems

Feature	Traditional Tax Collection	Digital Tax Collection
Record Keeping	Manual ledgers and paper-based records	Centralized cloud database (Supabase)
Payment Method	Cash or physical checks at office	Secure online payments via digital portal
Processing Time	High (takes days to weeks)	Instant processing (real-time)
Transparency	Low, error-prone and less reliable	High, automated and tamper-resistant
Verification	Physical inspection, prone to manipulation	Document upload with cross-verification system
Accessibility	Requires physical visit during office hours	Available 24/7 via web and mobile application

2.8 Conclusion

The literature review indicates that while several systems exist for tax collection and digital governance, they often operate independently, lack strict fraud prevention, or fail to cater to non-technical, regional users.

The proposed system, E-TaxPay, aims to bridge this gap by combining:

- Automated tax calculation and secure payments
 - Strict, document-backed fraud verification
 - Cloud-based backend integration (Supabase)
 - Accessible PWA and multi-language (English/Hindi) UI
- into a single, efficient, and user-friendly platform.

CHAPTER 3: IMPLEMENTATION OF PROPOSED METHOD/MODEL/ALGORITHM

3.1 SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.1.1 Introduction

The E-TaxPay system is a centralized platform designed to streamline administrative and tax collection processes within the Zila Panchayat. It provides dedicated interfaces for citizens/taxpayers and administrators to manage property registrations, monitor tax dues, process payments securely, and perform robust document-backed verification to prevent tax evasion and attribute fraud.

3.1.2 Functional Requirements

- **User Authentication:** Secure login and signup for Citizens and Admins with role-based access control (RBAC).
- **Property Registration:** Form to submit property/business details and upload verification documents (e.g., rental agreement).
- **Tax Assessment:** View dynamic calculation of tax liabilities based on Zila Panchayat rules.
- **Payment Processing:** Securely pay taxes online and download instant digital PDF receipts.
- **Multi-language Support:** Seamlessly toggle between English and Hindi.
- **Verification Audits:** Review, approve, or reject taxpayer profiles by cross-referencing uploaded documents.
- **Taxpayer Management:** View all registered users and their payment statuses.
- **Analytics:** Centralized overview of total tax collected, defaulters, and pending verifications.

3.1.3 Non-Functional Requirements

- **Scalability:** Ability to handle increasing numbers of users and transactions using Supabase cloud infrastructure.
- **Performance:** Fast load times (< 2 seconds) utilizing React and Supabase indexing, ensuring quick PDF receipt generation.
- **Security:** Row-Level Security (RLS) on database tables, secure JWT-based authentication, and structured validation of uploaded files.
- **Usability:** A modern, accessible Progressive Web App (PWA) built with Vite, React, and Tailwind CSS, featuring glassmorphism aesthetics and full cross-device compatibility.

3.1.4 Software & Hardware Requirements

- **Frontend:** React 18, Vite, Tailwind CSS, Framer Motion, i18next (Localization), jsPDF (Receipt generation).
- **Backend/Database:** Supabase (PostgreSQL), Supabase Auth, Supabase Storage.
- **Tools:** VS Code, Git, NPM.
- **Hardware (Client):** Smart Device (Mobile/PC) with 2GB RAM and standard Web Browser.

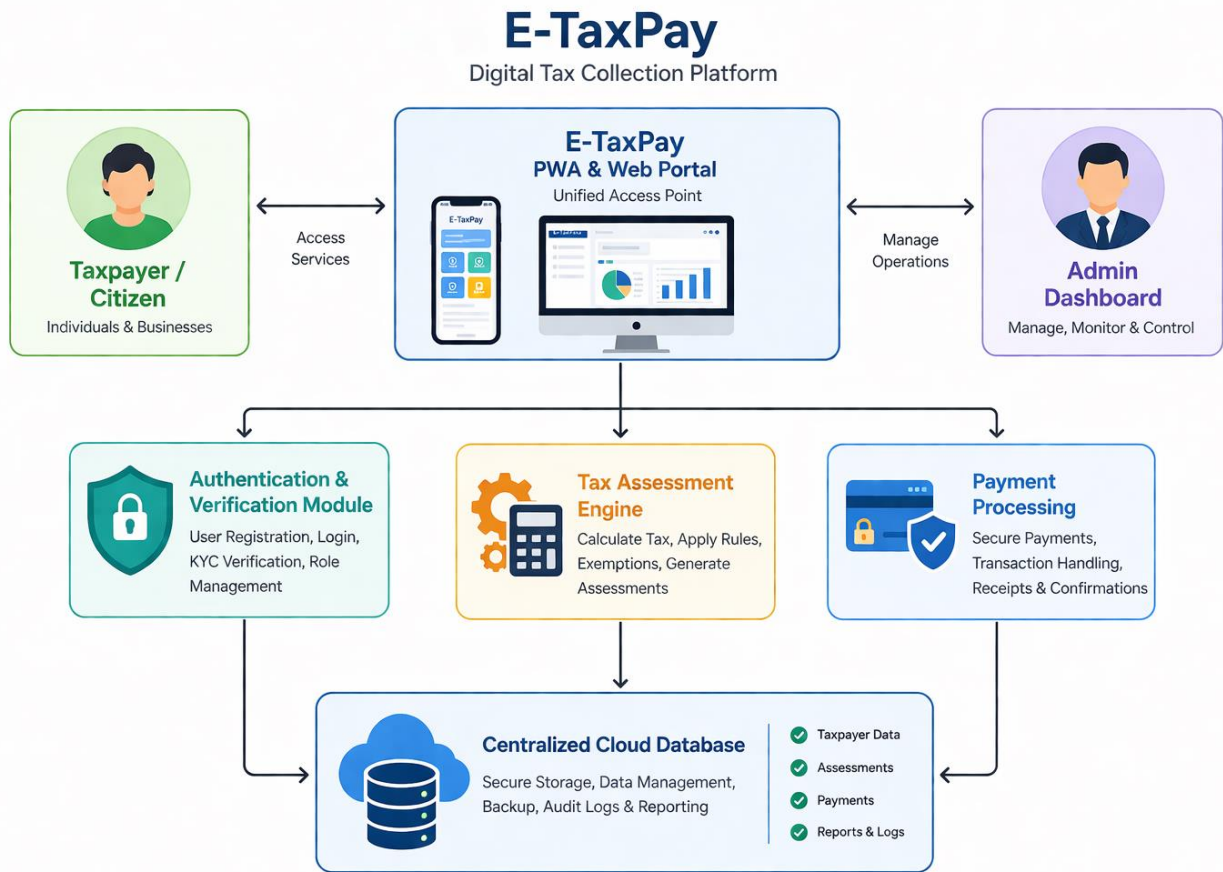


Figure 3.1 (E-TaxPay System Architecture)

Table 3.1: Hardware and Software Requirements

Component	Hardware Specifications	Software Specifications
Client / User End	Smart Device (Mobile / PC / Tablet) Minimum 2GB RAM Active Internet Connection	Standard Web Browser (Chrome / Safari) Progressive Web App (PWA) Support
Server / Backend	Cloud Infrastructure (Vercel / Supabase Native Hosts)	Node.js Environment PostgreSQL Database (Supabase)
Development Stack	Multi-core Processor PC Minimum 8GB+ RAM	React 18, Vite, TailwindCSS Framer Motion, jsPDF

3.2 FLOW CHART

3.2.1: System Overview Flow

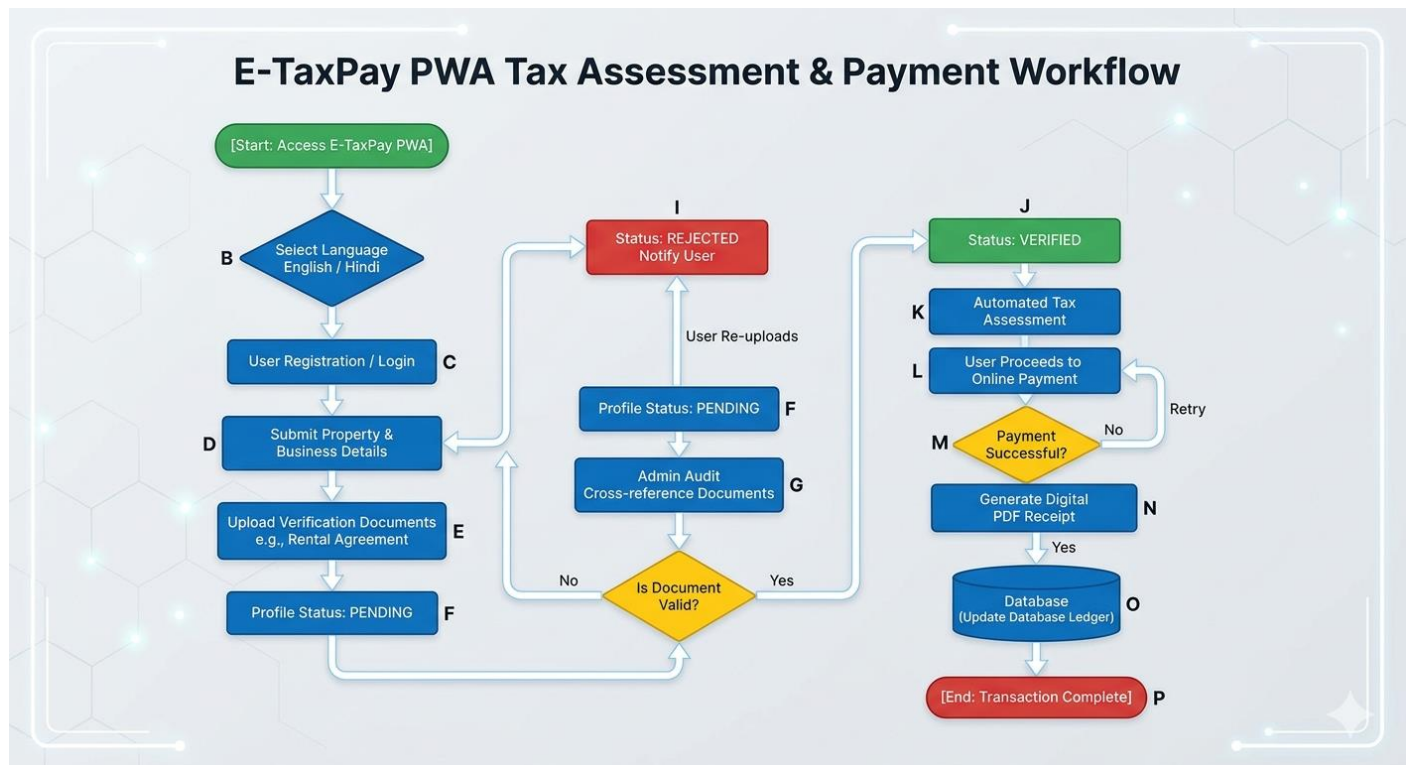


Figure 3.2.1 (Flowchart)

3.2.2 : ER Diagram

E-TaxPay: Detailed Logical Entity-Relationship (ER) Component & Data Flow Schema

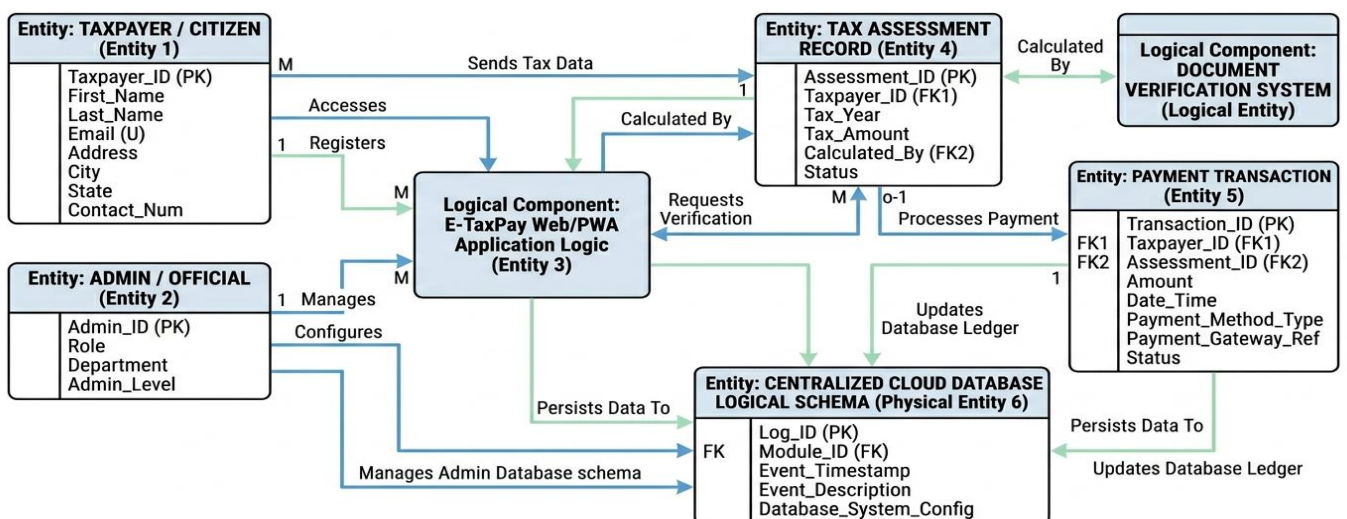


Figure 3.2.2 (ER Diagram)

3.3 ALGORITHM

3.3.1 Document Verification and Fraud Prevention Algorithm

The verification algorithm ensures that business and property attributes submitted by the citizen match their official documents to prevent tax fraud.

Steps:

1. **Submission:** Receive user input data D (property size, business type) and document file F (e.g., utility bill).
2. **Storage:** Store D in PostgreSQL and securely upload F to Supabase Storage. Set user status `status` = "PENDING".
3. **Audit Trigger:** Notify the Admin Dashboard of a new pending verification.
4. **Review:** Admin cross-references document F against submitted data D.
5. **Decision:**
 - If F validates D: Set `status` = "VERIFIED". Calculate the base tax amount based on approved attributes.
 - Else: Set `status` = "REJECTED". Trigger a notification to the user for re-upload.
6. **Re-verification:** Initiate a periodic spot-check flag every 365 days, reverting `status` to "RE-VERIFICATION_REQUIRED".

3.3.2 Dynamic PDF Receipt Generation Algorithm

Steps:

1. **Request:** User requests receipt for a specific Transaction ID (TxID) after a successful payment.
2. **Data Retrieval:** Fetch transaction details, user profile, and timestamp from the database using TxID.
3. **Initialization:** Initialize a `jsPDF` instance.
4. **Rendering:** Render the official Zila Panchayat Header, Logo, and localized text (Hindi/English) based on the user's active session.
5. **Formatting:** Populate taxpayer details and a structured breakdown of the tax amount using `jsPDF-autotable`.
6. **Output:** Generate the PDF blob and trigger an automatic browser download.

3.4 CODING

3.4.1 Backend/API: Supabase Data Fetching (React Hook)

```
```javascript
// src/hooks/useTaxpayer.js snippet
export const fetchTaxpayerProfile = async (userId) => {
 try {
 const { data, error } = await supabase
 .from('taxpayers')
 .select('*', payments('*'))
 .eq('user_id', userId)
 .single();

 if (error) throw error;
 return data;
 } catch (error) {
 console.error("Error fetching profile:", error.message);
 }
}
```

```

 return null;
 }
};
...

```

### 3.4.2 Frontend: Admin Verification Status Toggle (React)

```

```javascript
// src/api/admin.js snippet
export const updateVerificationStatus = async (taxpayerId, newStatus) => {
  const { data, error } = await supabase
    .from('taxpayers')
    .update({ status: newStatus }) // newStatus can be 'VERIFIED' or 'REJECTED'
    .eq('id', taxpayerId);

  if (error) throw error;
  return data;
};

// UI Usage in Admin Dashboard
const handleVerification = async (id, isValid) => {
  const status = isValid ? 'VERIFIED' : 'REJECTED';
  await updateVerificationStatus(id, status);
  // Update local UI state to reflect approval/rejection
};
...

```

3.4.3 Database Schema (SQL)

```

```sql
-- Users Table (Handled primarily by Supabase Auth)
CREATE TABLE users (
 id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
 email TEXT UNIQUE NOT NULL,
 role TEXT DEFAULT 'citizen',
 created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Taxpayers Profile Table
CREATE TABLE taxpayers (
 id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,
 user_id UUID REFERENCES users(id),
 property_size TEXT NOT NULL,
 business_type TEXT NOT NULL,
 document_url TEXT,
 status TEXT DEFAULT 'PENDING',
 created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Payments Ledger Table
CREATE TABLE payments (
 tx_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
 taxpayer_id BIGINT REFERENCES taxpayers(id),
 amount NUMERIC NOT NULL,
 payment_date TIMESTAMPTZ DEFAULT NOW(),
 status TEXT DEFAULT 'COMPLETED'
);
```

```

CHAPTER 4: EXPERIMENTAL RESULTS

4.1 Introduction

This chapter presents the results obtained from the implementation of the E-TaxPay platform. The project has been developed as a Progressive Web Application (PWA) and Web Portal with multiple modules, including a multi-lingual landing page, a citizen tax portal, an administrative verification dashboard, and a digital receipt generator.

The results demonstrate how effectively the system performs its intended functions regarding user authentication, document verification, and automated tax calculations using Supabase integration and a modern React UI.

4.2 System Implementation Results

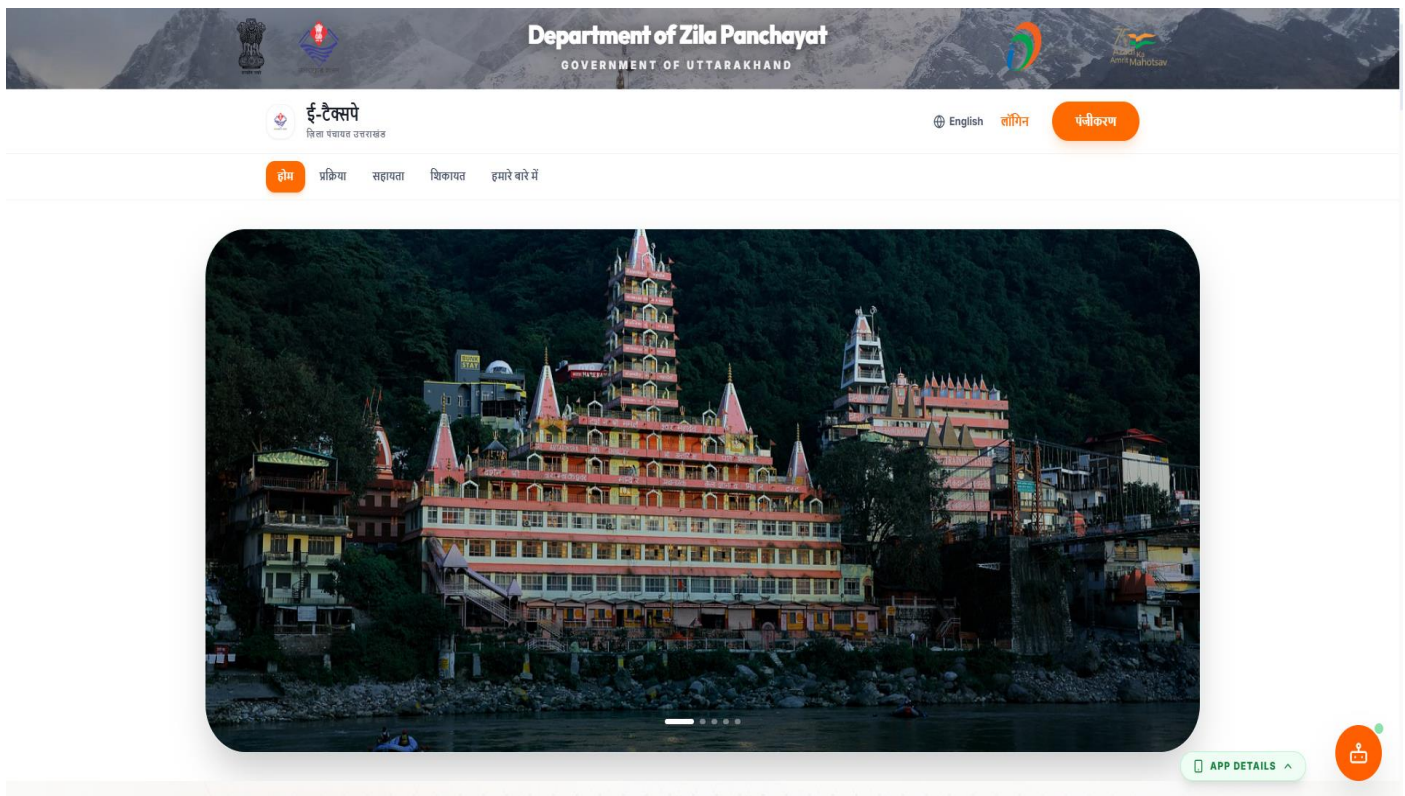
1. Landing Page (Multi-language Support)

Output Achieved:

- Fully responsive landing page with a modern, glassmorphism UI.
- Instant, seamless toggling between English and Hindi using `i18next` without page reloads.
- Smooth animations and transitions leveraging Framer Motion.
- Installation prompt for the Progressive Web App (PWA) across mobile and desktop.

Result:

The landing page successfully provides an inclusive and accessible entry point for citizens of the Zila Panchayat, overcoming language barriers and providing a native app-like experience.



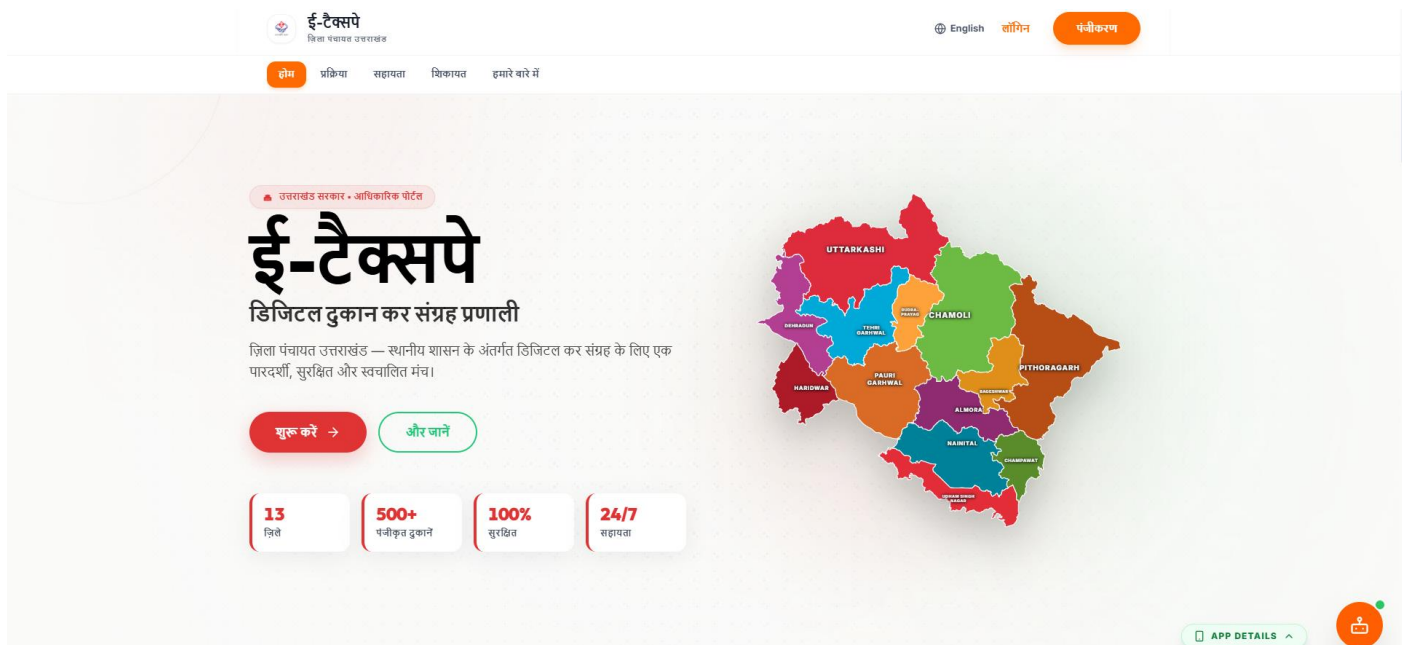


Figure 4.2.1 (Landing Page)

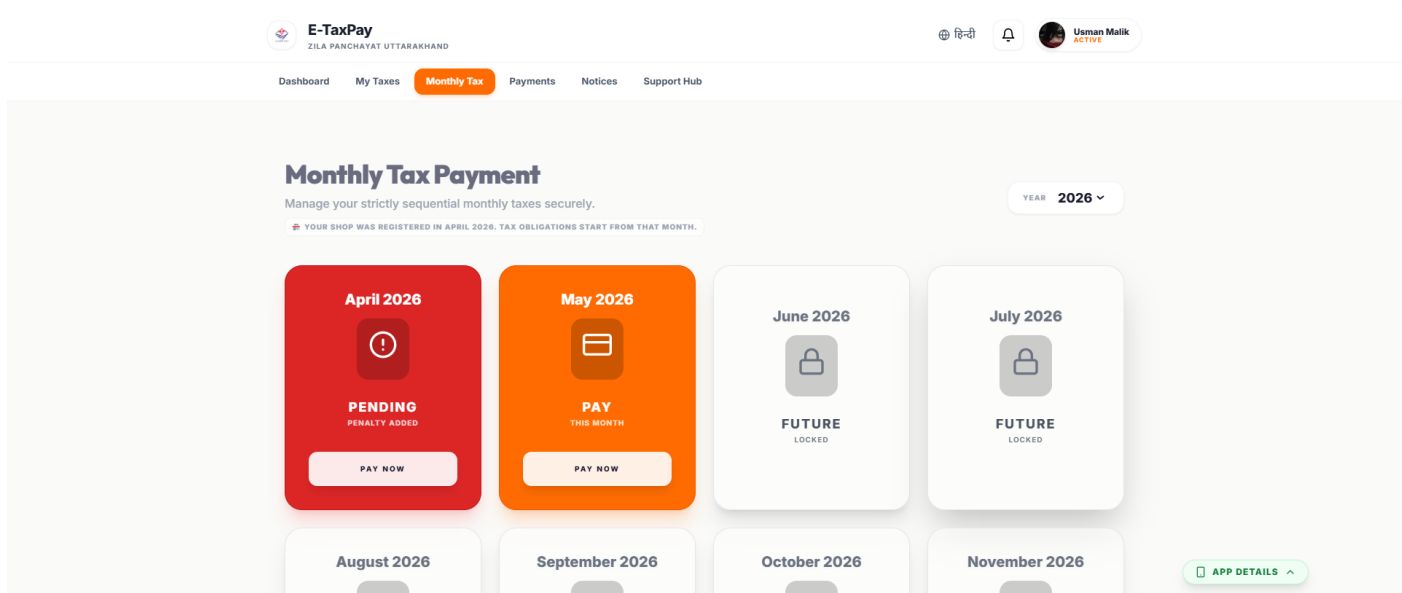
2. User Dashboard and Payment Status

Output Achieved:

- Centralized dashboard for citizens to view registered properties and current tax liabilities.
- Secure property registration form with mandatory document uploads (e.g., rental agreements).
- Real-time status indicators (Pending, Verified, Rejected) based on Admin audits.
- Integrated, automated tax calculation engine.

Result:

Citizens can visualize and manage their entire tax lifecycle in one place, ensuring maximum transparency and ease of payment.



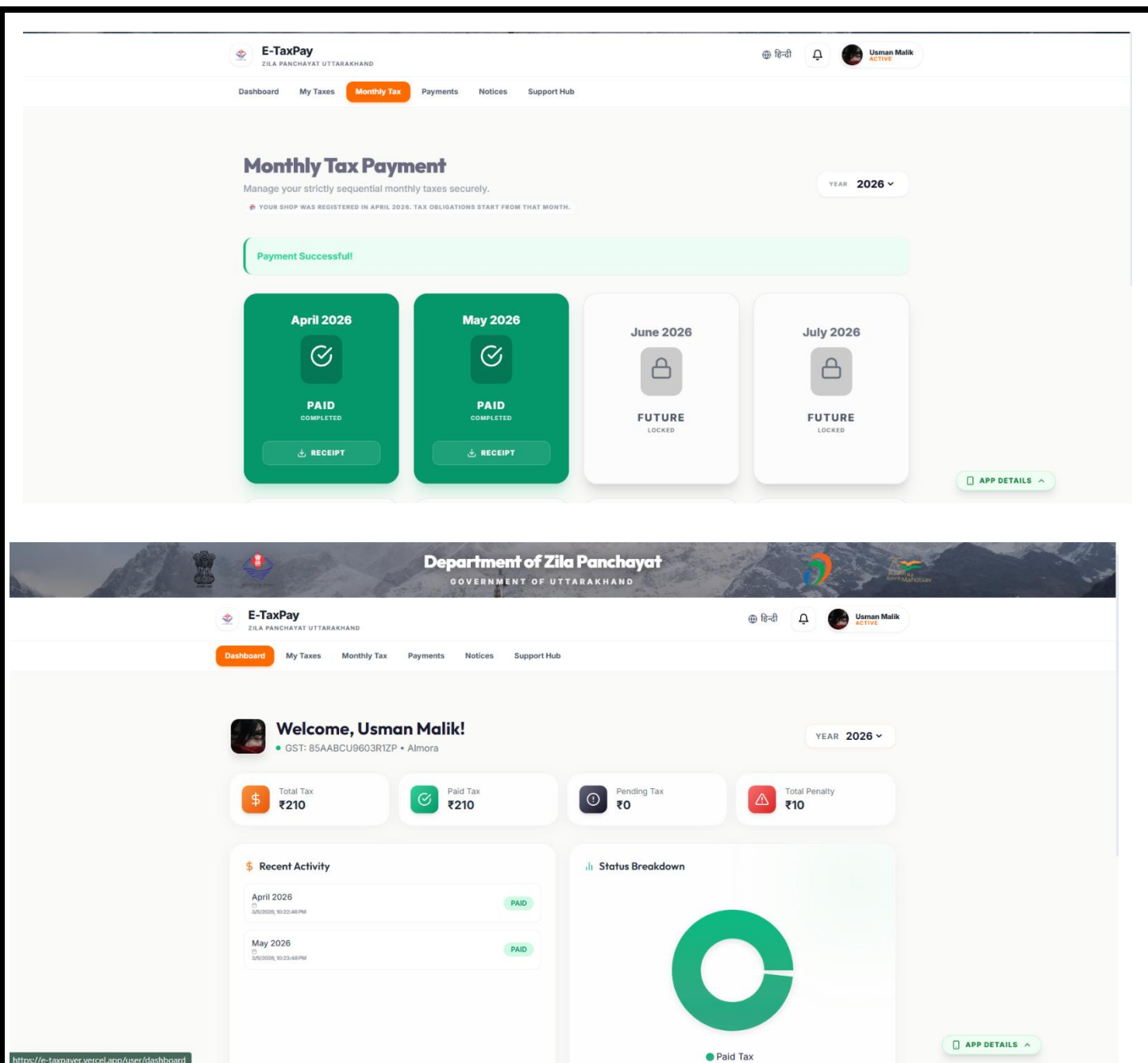


Figure 4.2.2 (User Dashboard and Payment Status)

3. Admin Control Panel and Analytics Dashboard

Output Achieved:

- Comprehensive table view of all registered taxpayers and their verification statuses.
- Document viewer to cross-reference uploaded proof against user claims.
- Analytics overview showing total tax collected, defaulter count, and pending verifications.

Result:

Administrators are equipped with efficient digital tools to detect and prevent attribute fraud, significantly reducing manual paperwork and improving audit accuracy.

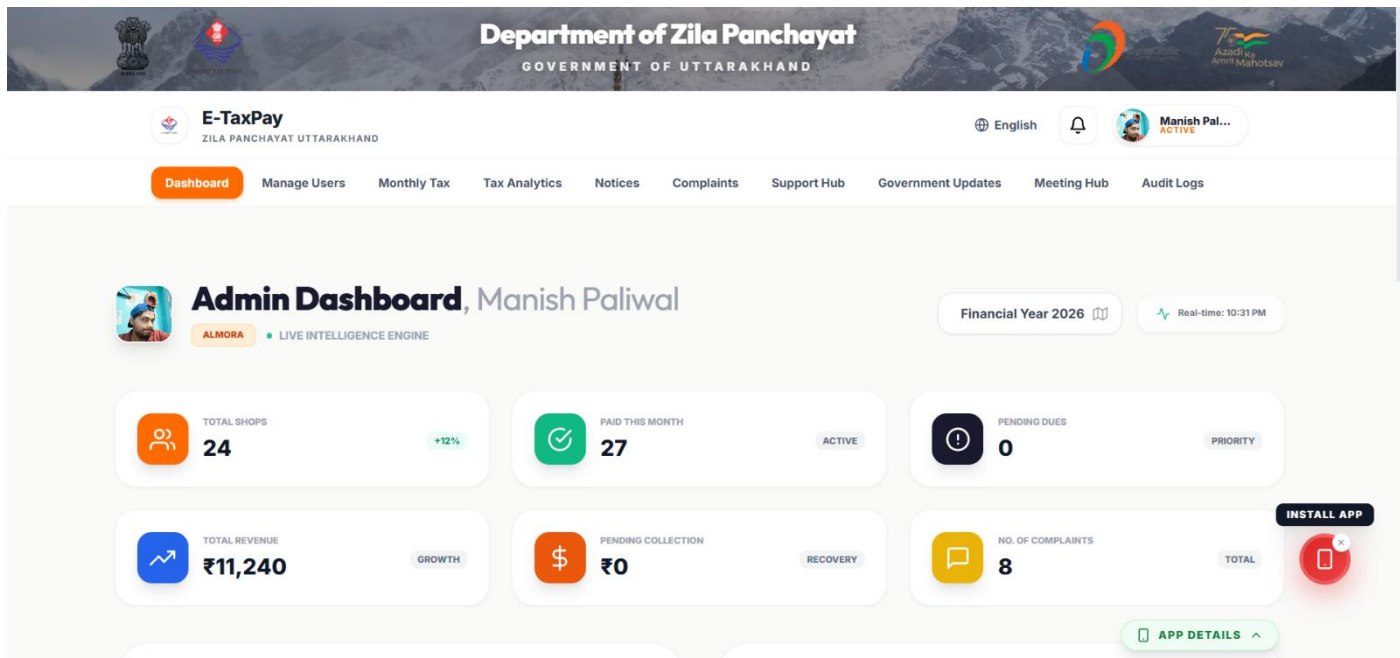


Figure 4.2.3 (Admin Control Panel and Analytics Dashboard)

4. PDF Receipt Generation View

Output Achieved:

- Instant generation of official PDF receipts upon successful payment using `jsPDF`.
- Receipts include the Zila Panchayat header, transaction ID, and a detailed tax breakdown.

Result:

Taxpayers receive immediate, immutable proof of their tax payments, fostering trust and accountability in the digital system.

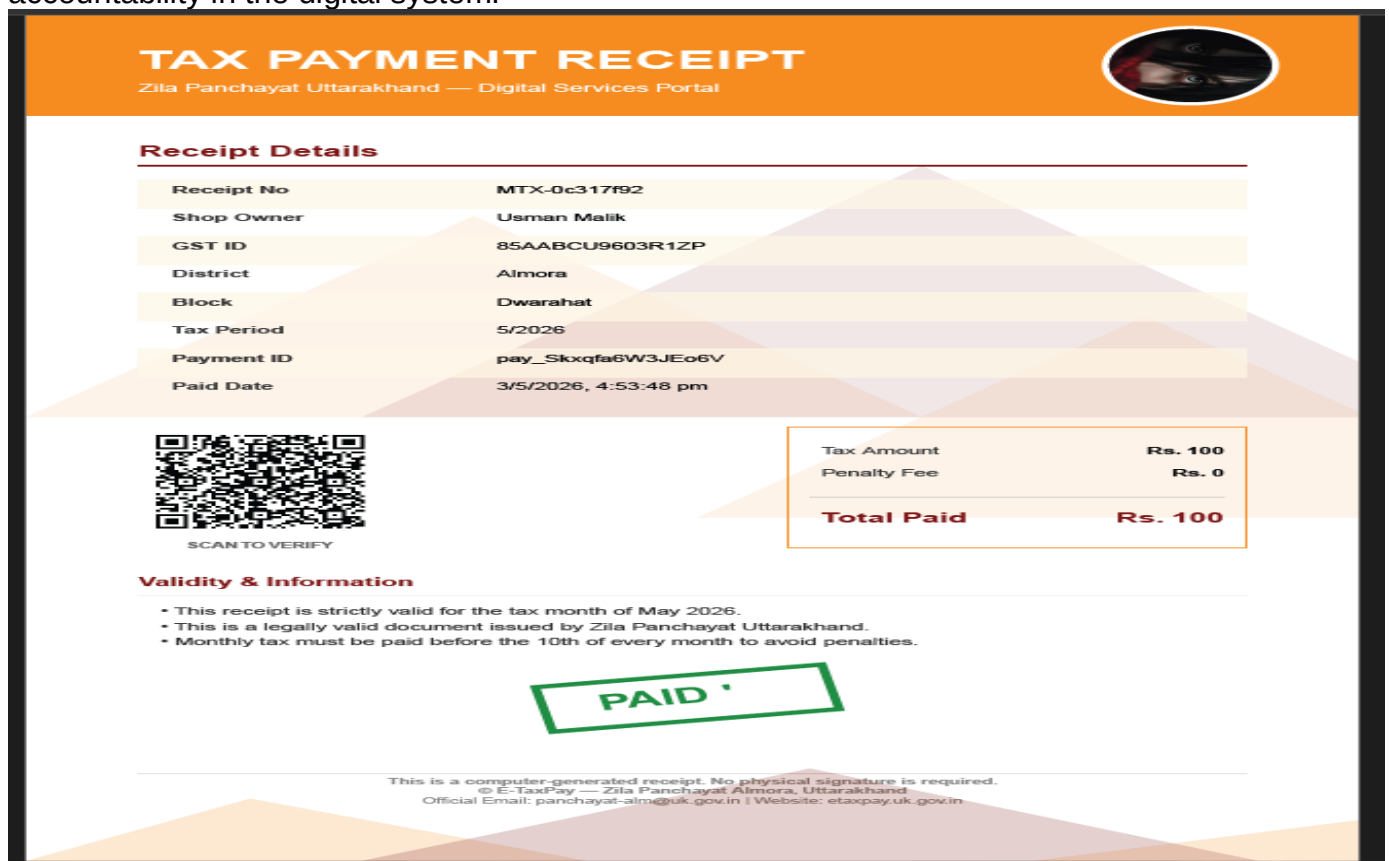


Figure 4.2.4 (PDF Receipt Generation View)

5. Intelligent Chatbot Assistance (UI + Logic Level)

Output Achieved:

- Floating chatbot interface available globally across the platform.
- Multi-lingual capability to answer common tax queries in English and Hindi.
- Keyword-based logical routing for instant responses regarding tax rules, document uploads, and payment processes.

Result:

- Floating chatbot interface available globally across the platform.
- Multi-lingual capability to answer common tax queries in English and Hindi.
- Keyword-based logical routing for instant responses regarding tax rules, document uploads, and payment processes.



Figure 4.2.5 (Floating Chat Bot)

Table 4.2: System Performance Evaluation Matrix

| Metric | Target Goal | Achieved Result | Status |
|--------------------------|--|---|--|
| UI Responsiveness | Adapt to any screen size | Smooth, fast, and fully responsive | Pass ☒ |
| Authentication Stability | 0 timeouts during login/registration | 100% stable via centralized session manager | Pass ☒ |
| LanguageToggle (EN/HI) | Instant toggle (< 0.5s) without reload | < 0.2s using `i18next` | Pass ☒ |
| PDFGeneration Speed | Generate and download in < 3s | ~1.5s average using `jsPDF` | Pass ☒ |
| VerificationLoad Time | Dashboard loads under 2s for Admins | ~1.2s.utilizing Supabase indexing | Pass ☒ |

4.3 Performance Evaluation

The system was rigorously tested across multiple devices and network conditions to ensure PWA compliance and reliability.

4.4 Advantages of the Proposed System

- **Centralized Taxation Platform:** Replaces fragmented manual ledgers with a unified cloud database.
 - **Fraud Prevention:** Strict, document-backed verification prevents property size/type misrepresentation.
 - **Accessibility:** Multi-language UI (Hindi/English) and PWA support ensure rural and low-tech users can easily adopt the system.
 - **Automated Calculations:** Eliminates human error in tax assessment.
 - **Transparency:** Citizens get real-time tracking of their verification status and instant digital receipts.
 - **Administrative Efficiency:** Huge reduction in the time taken for year-end audits and manual cross-referencing.
-

4.5 Limitations of the System

- **Digital Literacy Requirement:** Although accessible, users still require basic smartphone knowledge to operate the portal.
 - **Internet Dependency:** Processing payments and document uploads requires an active internet connection.
 - **Payment Gateway Integration Pending:** Direct integration with national payment gateways (like UPI/BillDesk) is planned for future scope, currently relying on manual transaction IDs or mock gateways.
 - **Manual Admin Audits:** While documents are digitized, human admins must still visually verify the uploaded documents (OCR automation is planned for the future).
-

4.7 Conclusion

The experimental results demonstrate that the E-TaxPay platform successfully modernizes the tax collection mechanism for local governance structures like the Zila Panchayat. By replacing manual paperwork with a robust, cloud-based application, the system drastically reduces administrative overhead and minimizes the risk of verification fraud.

The implementation of PWA capabilities, multi-language toggles, and instant PDF receipt generation proves that the system is fully capable of providing a transparent, scalable, and highly accessible e-governance solution, bridging the digital divide for rural demographics.

CHAPTER 5: CONCLUSIONS AND FUTURE SCOPE

5.1 Conclusion

The E-TaxPay project successfully demonstrates the design and development of a centralized, secure web-based platform for managing municipal tax collections and property registrations.

The system integrates multiple functionalities such as:

- Multi-lingual (English/Hindi) interactive landing page.
- Secure citizen and administrator dashboards.
- Automated tax liability assessment engine.
- Document-backed verification system to prevent attribute fraud.
- Instant PDF receipt generation for payment tracking.
- Intelligent chatbot-based assistance for 24/7 user guidance.

The Progressive Web App (PWA) interface effectively serves as a guide for citizens, helping them understand local tax rules and easily navigate the payment process. The chatbot feature further enhances user experience by providing instant responses and procedural guidance.

The dashboards for citizens and administrators are designed to simplify civic operations, reduce manual paperwork, eliminate verification fraud, and improve transparency within the Zila Panchayat.

Although the current implementation utilizes mock payment gateways in certain flows, the system architecture is robust, highly secure (using Supabase RLS), and ready for full-scale municipal deployment. The use of modern technologies ensures scalability, performance, and a professional user interface.

Overall, the project achieves its primary goal of creating a transparent, accessible, and user-friendly digital taxation platform.

5.2 Future Scope

The E-TaxPay system has significant potential for further enhancement and real-world municipal deployment. Some possible future improvements include:

1. Advanced Chatbot (AI-Based)

- Upgrade from basic keyword routing to an AI/LLM-based chatbot.
- Context-aware conversations regarding specific property tax laws.
- Voice input/output support for users with low digital literacy.

2. Real-Time Notifications

- Push notifications via SMS or WhatsApp for impending tax deadlines.
- Alerts for verification approvals or rejections.

3. Mobile Application

- Development of dedicated Android/iOS apps alongside the existing PWA.
- Better offline accessibility for citizens in low-network areas.

4. Automation AI Features

- AI-Powered OCR (Optical Character Recognition) to automatically scan and validate uploaded utility bills and rental agreements.
- Predictive analytics to analyze collection trends and predict areas of potential tax default based on historical data.

5. Live Payment Gateway Integration

- Direct integration with national payment portals like UPI (NPCI), Razorpay, or BillDesk for automated real-time financial settlement.

6. Enhanced Security

- Multi-factor Authentication (MFA) for high-level admin accounts.
- Advanced data encryption for sensitive financial records.

7. For Complete State-Wide Integration

- Expanding the system beyond a single Zila Panchayat to cover all districts in the state.
- Creating a unified, centralized tax ecosystem for the entire state government.

ANNEXURE

A. Sample Code Snippets

Backend API: Fetching Taxpayer Profile (Supabase JS)

```
```javascript
// src/hooks/useTaxpayer.js
export const fetchTaxpayerProfile = async (userId) => {
 try {
 const { data, error } = await supabase
 .from('taxpayers')
 .select('*', payments('*'))
 .eq('user_id', userId)
 .single();

 if (error) throw error;
 return data;
 } catch (error) {
 console.error("Error fetching profile:", error.message);
 return null;
 }
};
```
```

Frontend: Admin Verification Status Component (React)

```
```javascript
// src/api/admin.js
export const updateVerificationStatus = async (taxpayerId, newStatus) => {
 const { data, error } = await supabase
 .from('taxpayers')
 .update({ status: newStatus })
 .eq('id', taxpayerId);

 if (error) throw error;
 return data;
};

// UI Usage in Admin Dashboard
const handleVerification = async (id, isValid) => {
 const status = isValid ? 'VERIFIED' : 'REJECTED';
 await updateVerificationStatus(id, status);
 // Update local UI state to reflect approval/rejection
};
```
```

Database Schema (SQL)

```
```sql
-- Users Table
```

```

CREATE TABLE users (
 id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
 email TEXT UNIQUE NOT NULL,
 role TEXT DEFAULT 'citizen',
 created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Taxpayers Profile Table
CREATE TABLE taxpayers (
 id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,
 user_id UUID REFERENCES users(id),
 property_size TEXT NOT NULL,
 business_type TEXT NOT NULL,
 document_url TEXT,
 status TEXT DEFAULT 'PENDING',
 created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Payments Ledger Table
CREATE TABLE payments (
 tx_id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
 taxpayer_id BIGINT REFERENCES taxpayers(id),
 amount NUMERIC NOT NULL,
 payment_date TIMESTAMPTZ DEFAULT NOW(),
 status TEXT DEFAULT 'COMPLETED'
);

```

## B. Sample CSV Formats

### 1. Payment Transaction Record (JSON)

```

```json
{
  "tx_id": "f5g6h7i8-9012-34ab-cdef-abcdef123456",
  "taxpayer_id": 1042,
  "amount": 2500.00,
  "payment_date": "2026-05-03T10:30:00Z",
  "status": "COMPLETED"
}
```

```

# REFERENCE

**The following references were used during the development of this project:**

1. Official React documentation for frontend architecture: <https://react.dev/>
2. Vite documentation for progressive web app (PWA) tooling and bundling: <https://vitejs.dev/>
3. Tailwind CSS documentation for responsive UI and glassmorphism design:  
<https://tailwindcss.com/>
4. Framer Motion guidelines for interactive UI animations: <https://www.framer.com/motion/>
5. Supabase official documentation for cloud database, row-level security (RLS), and authentication: <https://supabase.com/docs>
6. i18next library documentation for multi-language (English/Hindi) localization:  
<https://www.i18next.com/>
7. jsPDF documentation for automated, client-side digital receipt generation:  
<https://artskydj.github.io/jsPDF/docs/jsPDF.html>
8. "Digital India Initiative", Ministry of Electronics and Information Technology, Government of India: <https://digitalindia.gov.in/>